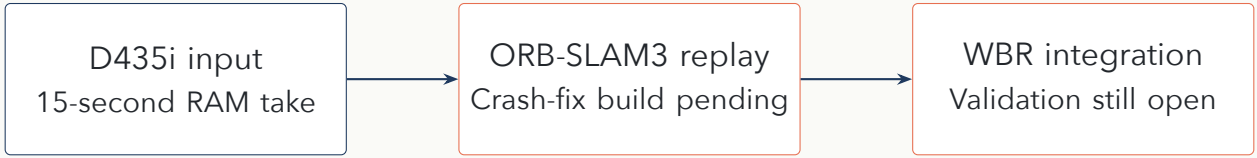
Known-data accuracy is promising; robot readiness remains open.

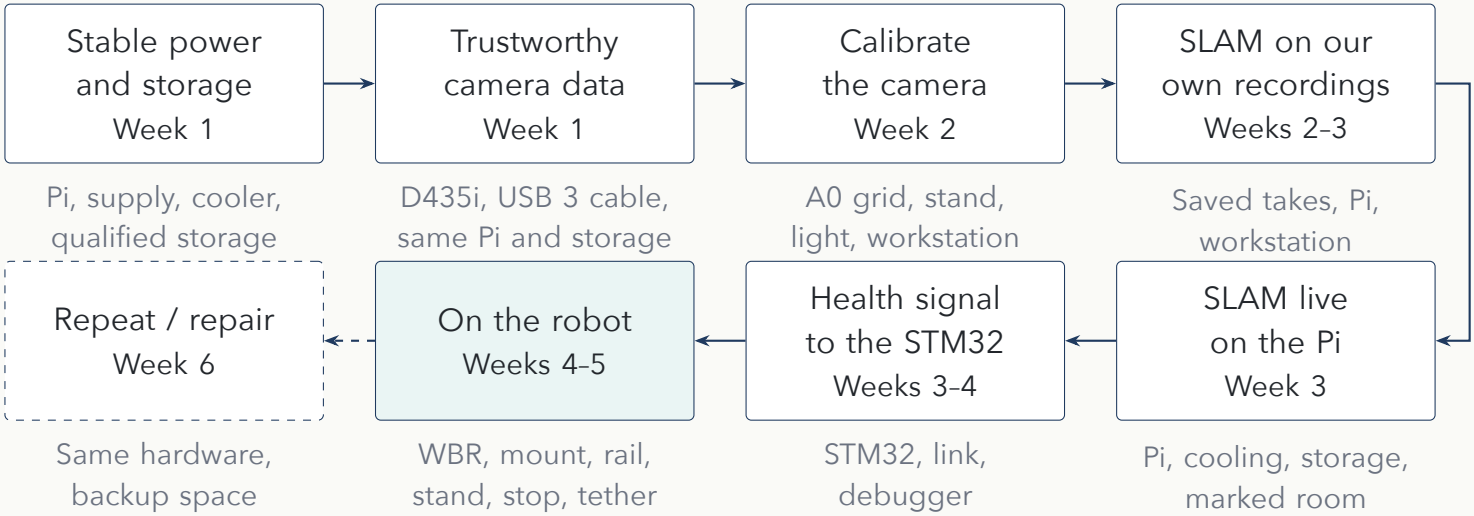
WBR means wheel-biped robot. Snapshot: 14 October 2026.



- Three runs of EuRoC MH01, a dataset with a reference trajectory, measured 3.7–4.2 cm trajectory error; under-voltage invalidates their speed result.
- The 8 GB Pi boots, but supply, large-data storage and D435i calibration are unresolved.
- STM32H7 retains body estimation, balance, motor and jump safety.

First deployment: optional position information during supervised slow motion.

The whole road: today to a supervised robot trial



Conditional weeks from 14 October. A failed step moves the schedule; it never shortens a test.

A few terms make the plan easier to read

Term	Meaning in this roadmap
SLAM	Simultaneous localization and mapping: estimate camera position while building a map.
VIO / IMU	Visual-inertial odometry combines images with an inertial measurement unit: gyroscope and accelerometer.
Calibration	Measure camera geometry, camera–IMU alignment, timing offset and sensor noise.
ROS / bag / replay	Robot Operating System passes messages; a bag records them; replay feeds a saved recording back into software.
Pose / frame / reset	Pose is position and orientation; frame defines axes and origin; reset starts a new estimate.
Health / optional input	Health reports valid, lost or old estimates. Optional position information never owns balance or motor safety.

Stable power and storage

Preparation; platform check N0

What	Make the next lab visit usable.
How	Confirm parts, qualify the data path, prepare the target and prove file conversion on synthetic data.
Tools	Power/storage checks; offline image and IMU conversion; checksums for exact-copy verification.
Hardware	8 GB Pi, 5.1 V/5 A supply, cooler, storage, local console; camera stays idle.
Done when	Named owners for every blocker; conversion preserves pixels, counts, units and timestamps on known samples.
Time	2–5 engineer-days; parts and workstation waits are additional.

Resolve these dependencies before camera time

Dependency	Current state	Next owner action
Power / access	5 V/3 A still under-voltages; ordered supply arrival unknown. Wi-Fi unverified; Ethernet loose.	Confirm suitable supply/cooling and local console or verified network access.
ROS / storage	Camera wrapper not installed. USB stick measured 1.09 MB/s.	Review installation separately; borrow or buy and qualify large-bag storage.
Calibration tools	ROS 1 bag conversion and long stationary recorder are untested.	Prove conversion offline; confirm workstation and recorder stop/recovery.
Target / people	Measured target, overnight permission and controller owner unconfirmed.	Prepare print; name controller engineer; request conditional lab slots.

The owner must explicitly authorize camera work before any capture.

Trustworthy camera data

Input checks N0 → N1 → N2 → N3

What	Prove the same camera can stream, record and replay reliably.
How	Check platform, USB and raw data in order; then record ROS messages and compare full replay.
Tools	Guarded sensor checks; reviewed ROS 2 Jazzy camera driver; recording/replay comparison.
Hardware	Same Pi, stable supply/cooler, D435i, direct USB 3 cable and qualified bag storage.
Done when	USB reports at least 5000M; all three raw tests meet the next page; recorded and replayed counts/fields agree.
Time	0.5–3 engineer-days; reserve 3–4 h, with diagnosis allowed to use the slot.

What a trustworthy raw recording looks like

Raw-data check N2; ROS record/replay N3 follows

Check	Required visible result
Exact raw profile	IR1: Y8 grayscale, 640×480 at 15 Hz; gyroscope 200 Hz; accelerometer 100 Hz.
Three stages	20 seconds, then 105 seconds, then the mandatory 300-second soak.
Measured rates	Image 14.25–15.75 Hz; gyro 190–210 Hz; accel 95–105 Hz.
Gaps and order	Image gaps ≤0.20 s; motion gaps ≤0.05 s; increasing timestamps/frame numbers; no missing image frames.
Health and stop	All original startup, boundary, content, kernel, power and normal-stop checks reviewed; raw logs retained.

> A short successful take does not replace the five-minute soak.

Calibrate the camera: collect usable measurements

Calibration acquisition within N4A / N4B

What	Measure this unit rather than rely on example defaults.
How	Record three target-motion takes; separate room trials and stationary noise data; inspect the first take.
Tools	Reviewed stereo recorder; future long-duration IMU recorder; verified lossless export.
Hardware	Same camera/Pi, measured A0 AprilGrid (tag-pattern target), stand, light, calipers and backup space.
Done when	Three 120 s target takes and separate 90 s room trials have usable images and ordered timing; two copies match.
Time	1–2 engineer-days; 3–4 h target slot; separately 30 min warm-up + 15–24 h stationary data.

SLAM on our own recordings

Known reference N4A; camera replay N4B

What	Check repeatable output before going live.
How	Finish the existing crash-fix build; rerun the known EuRoC dataset; calibrate and replay three own-camera takes plus unused evaluation data.
Tools	Native ORB-SLAM3; Kalibr alignment/timing solver; Allan noise analysis; trajectory plots.
Hardware	8 GB Pi with stable power; confirmed ROS 1/Kalibr workstation; immutable saved takes.
Done when	Build loads; three replay results show initialization, tracking losses and resets; settings and calibration have traceable reports.
Time	2–4 engineer-days; add 2–5 days if import/tool compatibility fails.

SLAM live on the Pi

Live estimator N4C; sustained resources N5

What	See whether current camera position arrives in time.
How	Hold settings fixed; run three repeats and a proposed 30-minute load with growing map, revisits and recovery.
Tools	ORB without a display; estimate-age, queue, drop/reset, processor/memory, temperature and power plots.
Hardware	Final Pi/camera/cooling/data path, textured marked route, operator and reviewer.
Done when	Full logs account for inputs and outputs; queues do not grow; age/resource plots stay within limits approved before testing.
Time	1–3 engineer-days; book 2–3 h after recorded-data review.

Measure how old an estimate is

- Estimate age is the time from image capture to the position result being available.
- First prove how camera and Pi clocks relate. If that relation is unknown, age is unknown.
- Report the median and the 95th/99th percentiles: values below which 50/95/99% of samples fall.
- Count every missing input, queued frame, tracking loss and reset alongside CPU, memory, heat and power.

> A fresh status message cannot make an old camera estimate fresh.

Algorithm comparison: keep one default, test one challenger

Scores from the independent review, checked against the algorithm review; judgments, not Pi benchmarks.

Candidate	Demo	CPU	Motion	/100	Decision
ORB-SLAM3 stereo/IMU	5	3	3	68	Default; existing tools
RTAB-Map own motion estimate	3	2	3	68	For an obstacle map
OpenVINS	3	4	3	61	First matched-data challenger
Wheels/body + depth	2	5	2	56	Requires working robot exports
ORB RGB-D	3	3	2	52	New input path needed
Basalt	2	3	3	53	Prove build/adapter benefit
VINS-Fusion	2	3	3	47	Prove port benefit
Kimera-VIO	2	2	3	48	More dependencies; defer

1–5: higher = better fit. RGB-D = color + depth. Total weights: demo 30%, CPU 20%, motion 20%, software support 15%, navigation 10%, licence 5%.

What would make us change the algorithm?

- Keep ORB-SLAM3 for the first position-estimation trial: build and input work already exist, but live Pi performance is still open.
- If trusted data and correct calibration still miss approved age or continuity limits, spend at most 2–3 days diagnosing; compare OpenVINS on identical retained recordings.
- Switch only if the challenger meets the same approved limits and its build, input conversion and failure behavior are verified.
- If the first goal becomes an obstacle/occupancy map, compare RTAB-Map under the complete Pi workload.
- A demonstrated wheel/body interface could make a smaller rolling demo faster; it changes the deliverable and does not prove camera VIO.

Health signal to the STM32

Receiver review and acceptance N6

What	Tell the controller when optional position information is usable.
How	Review the link, clocks, axes and invalidation rules; test missing, frozen, reset and corrupt input on synthetic cases.
Tools	Proposed bounded receiver; expected-versus-actual traces; local timeout on the STM32.
Hardware	Named controller engineer, confirmed STM32H7 board, reviewed link and debugger.
Done when	Traces show old/wrong/corrupt input rejected and optional input withdrawn after Pi/link loss within approved limits.
Time	2–4 engineer-days; design may overlap after review; acceptance follows sustained Pi tests.

On the robot: mounting and supported tests

Supported integration N7

What	Verify power, mounting and controller independence.
How	Review mount/axes and power rail; test with motors disabled, then separately approved supported faults.
Tools	Reviewed receiver; power, controller-loop and estimate-age traces; controlled fault tests.
Hardware	WBR, controller owner, battery/converter/fuse/link, mount/cooler, rated stand and independent stop.
Done when	Logs show no unexpected actuation/reset or control-loop violation; each fault removes optional input within approved limits.
Time	1–3 engineer-days; book 3–4 h after receiver review.

On the robot: supervised slow motion

Separately authorized motion N7

What	Repeat a slow trial while local balance remains independent.
How	Baseline the controller; use approved tethered trials and bounded manual room motion; test Pi loss and repeat startup/stop.
Tools	Operator runbook; checksummed utility bundle; Pi and STM32 traces.
Hardware	Approved tether/cleared route, local stop, controller owner, operator and spotter.
Done when	Retained trials meet agreed motion/stop limits; Pi loss is handled; another operator reproduces startup, stop and rollback.
Time	1–4 engineer-days; 2–3 h motion plus a separate repeat/repair slot.

Hardware and lab bookings: bench and calibration

Item	Phase	Status	Book when	Prepare first
8 GB Pi + D435i	Input → live	Have per notes; confirm loan	W1; retain through W6	Same unit, console, USB 3 cable
5.1 V/5 A supply + cooler	All Pi work	Supply ordered; cooler confirm with lab	Before first camera slot	Confirm arrival and power checks
Large-data storage	Input → live	Borrow / buy; unqualified	Before W1 input slot	Sustained-write and replay checks
128 GB microSD	Calibration only	Ordered; arrival unknown	Before target slot	Confirm exception; qualify short takes
A0 grid, stand, light, calipers	Calibration	Print/buy; tools confirm with lab	W2: 3–4 h	Measure grid, prove import route
Quiet bench + workstation	Calibration	Lab; confirm access	W2: 30 min + 15–24 h	Test IMU recorder, backup and stop

Hardware and lab bookings: live Pi and robot

Item	Phase	Status	Book when	Prepare first
Marked textured room	Live Pi	Lab; confirm route	W3: 2–3 h	Replay accepted; age/resource limits
STM32H7 + link + debugger	Health signal	Lab; board/owner unknown	W3–4; after review	Confirm board, voltage, pins and link
WBR + mount + battery rail	Supported robot	Lab; stock/design unknown	W4: 3–4 h	Review mount/power; receiver tests
Rated stand + independent stop	Supported robot	Lab; confirm suitability	With controller owner in W4	Approve fault procedure
Tether + clear route + spotter	Slow robot motion	Lab; confirm suitability	W5: 2–3 h	Supported tests; separate motion approval
Repeat/repair access + backups	Handoff	Lab; confirm access	W6: 2–4 h reserve	Preserve failures; list unresolved work

Software stack by phase

Phase	Software / purpose	What must be ready
Preparation / input	Ubuntu + ROS 2 Jazzy; RealSense SDK/driver; existing raw checks; rosbag2 recorder/replay.	Review wrapper install; verify actual profiles and complete replay.
Calibration	Lossless PNG + full-precision IMU export → ROS 1 bag; Kalibr; Allan sensor-noise analysis.	Conversion/workstation and long recorder are future reviewed tools.
Saved-data / live SLAM	Native ORB-SLAM3 stereo-inertial; existing adapter; trajectory and health/resource plots.	Finish existing fix; preserve versions, settings and raw evidence.
STM32 health	Small reviewed Pi health adapter + bounded STM32 receiver; synthetic rejection tests.	Message, link and limits need controller approval before implementation.
Robot / handoff	Same frozen stack; startup/stop/rollback runbook; checksummed runtime bundle.	No balance authority on Pi; reuse a validated controller link.

Defer extra ports, Pi graphics and navigation until needed.

Next six weeks: preparation through live Pi

Week	Owner / lab	Engineering work	Visible result
1: 14–20 Oct	Confirm parts, camera loan, storage, console and workstation; request 3–4 h bench.	Prove conversion offline; target/recorder preparation; fresh input checks after authorization.	Reviewed input/replay receipts or a precise blocker.
2: 21–27 Oct	Book target 3–4 h and separate overnight; same camera/Pi.	Three target takes, unused room takes, stationary data; first-take review and backups.	Usable captures, two matching copies; calibration reports or recapture decision.
3: 28 Oct–3 Nov	Workstation; reserve marked route 2–3 h; agree controller limits.	Stable-power reference; repeated own-data replay; live and sustained-load measurements.	Tracking/reset and estimate-age plots; no growing queue; accept or diagnose.

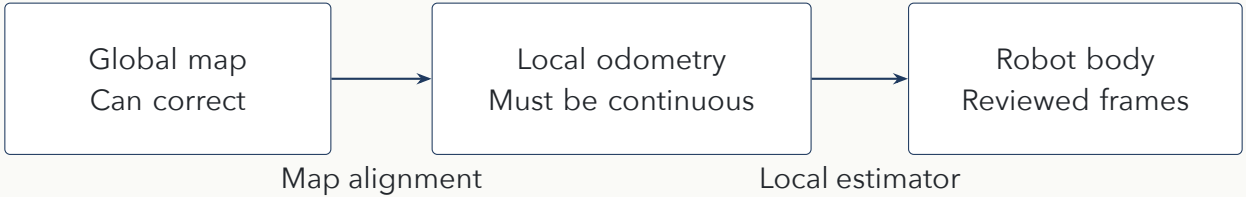
Dates are conditional targets. Unmet prerequisites carry forward; the soak and overnight recording remain unchanged.

Next six weeks: controller, robot and contingency

Week	Owner / lab	Engineering work	Visible result
4: 4–10 Nov	Controller owner; confirmed WBR, board, link, stand and stop; 3–4 h.	Accept receiver after live/resource review; motors-disabled then approved supported faults; mount/power checks.	Rejection/fallback traces and unchanged local control within approved limits.
5: 11–17 Nov	Separate motion approval; tether, route, operator/spotter; 2–3 h.	Supervised slow motion, Pi-loss trial, repeat startup/stop and review vibration/age.	Target: mounted tracking plus verified health/fallback behavior; no navigation claim.
6: 18–24 Nov	Hold 2–4 h repair/repeat slot; extend camera loan if needed.	Resolve the failing layer; freeze utility/settings bundle; second-operator repeat.	Reproduction report, verified backups and explicit remaining blockers.

If week 4 slips, finish supported tests before moving the robot.

Navigation is a later project.



- Validate acquisition-stamped wheel/body/contact telemetry, geometry, clocks and slip before fusion.
- Qualify obstacle coverage and occupancy mapping; sparse ORB landmarks are not an obstacle map.
- Nav2, a ROS navigation stack, needs reviewed continuous odometry, frames, stop bounds and controller veto.

D435i depth is a candidate obstacle sensor, not a qualified input.

The route traces to project evidence and primary sources.

- **Project state:** 13–14 October facts, N3 preparation, current lab runbook, calibration plan and N4A report.
- **Kalibr bag format** and recording conversion; noise guidance.
- **RealSense .db3 recording** uses ROS 2 rosbag2 format.
- **Raspberry Pi power guidance** and official supply specification.
- **Current cited ORB ROS wrapper** and OpenVINS Jazzy proposal.
- **REP-105**, Nav2 Jazzy odometry and RTAB-Map mapping.

Durations, later acceptance limits and switch rules are planning proposals. Hardware, calibration, estimate age and deployed performance remain unverified.